TOPORS-Related Retinopathy (Retinitis Pigmentosa 31, RP31): A Comprehensive Disease Characterization

Disease: TOPORS-Related Retinopathy · **Causal gene:** *TOPORS* (topoisomerase I-binding arginine/serine-rich protein, E3 ubiquitin/SUMO ligase; HGNC:21653; NCBI Gene 10210) · **MONDO:** MONDO:0700233 · **OMIM:** #609923 (RP31) · **Gene locus:** 9p21.1 · **Category:** Mendelian, autosomal dominant

Summary

TOPORS-Related Retinopathy is a rare, non-syndromic, adult-onset autosomal dominant rod–cone dystrophy (retinitis pigmentosa) caused by heterozygous truncating loss-of-function variants in the *TOPORS* gene at chromosome 9p21.1. The gene — not to be confused with the serial "retinitis pigmentosa 31 / RP31" naming — encodes a 1045-amino-acid RING-domain dual ubiquitin/SUMO1 E3 ligase. Disease arises through **haploinsufficiency**: the absence of mutant protein in patients, combined with the ubiquitous expression of *TOPORS*, favors a reduced-dosage mechanism over a dominant-negative one. TOPORS mutations account for approximately **1% of all autosomal dominant retinitis pigmentosa (adRP)** (PMID: 17924349, PMID: 18509552).

Mechanistically, TOPORS is a **component of the photoreceptor sensory (connecting) cilium**, localizing to basal bodies and centrosomes. Its loss impairs ciliary trafficking and the formation of photoreceptor outer segments, as demonstrated by zebrafish morpholino knockdown that produces defective retinal development rescuable by human *TOPORS* mRNA (PMID: 21159800). A parallel retinal mechanism involves **ubiquitin-proteasome proteostasis**: a retinal yeast two-hybrid screen identified the 26S proteasome subunit PSMC1/P26s4 as a TOPORS interactor localizing to the centrosome and the RPE/photoreceptor interface (PMID: 26872363). Clinically this produces the classic RP triad — night blindness, progressive peripheral visual-field constriction, and bone-spicule retinal pigmentation — sometimes preceded by a distinctive early perivascular cuff of RPE atrophy around the vascular arcades.

A striking feature of this gene is **dual inheritance**: while heterozygous truncating variants cause retina-only adRP, biallelic hypomorphic missense alleles (notably p.Pro10Gln, a Dominican founder allele) instead cause recessive **syndromic ciliopathies** — Joubert syndrome and oral-facial-digital syndrome (PMID: 37227088). This allele-specific, dosage-dependent pattern unifies TOPORS disease around primary-cilium dysfunction. There is **no approved disease-specific therapy** for RP31; management is supportive, with genetic counseling and treatment of complications such as cataract and cystoid macular edema. Progression is slow but relentless, and nutritional supplementation (vitamin A / DHA) has only low-certainty evidence of benefit.

Key Findings

Finding 1 — *TOPORS* haploinsufficiency at 9p21.1 causes autosomal dominant RP (RP31)

TOPORS is a three-exon gene encoding a 1045-amino-acid RING-domain E3 ubiquitin/SUMO ligase located at chromosome **9p21.1**. It was identified as the RP31 gene by positional cloning. Heterozygous truncating mutations — an insertion in a French-Canadian family and a deletion in a German family, with later-described alleles such as p.Glu808X and p.Arg857GlyfsX9 — segregate with adRP. The authors concluded that **"The ubiquitous nature of TOPORS expression and a lack of mutant protein in patients are highly suggestive of haploinsufficiency, rather than a dominant negative effect"** (PMID: 17924349). A follow-up mutation screen established that **"Point mutations and small insertions or deletions in TOPORS cause approximately 1% of adRP"** (PMID: 18509552).

This finding anchors the disease: the causal gene is unambiguously *TOPORS*, the inheritance is autosomal dominant, the molecular lesion is loss of function, and the pathogenic mechanism is haploinsufficiency (reduced gene dosage below a functional threshold in photoreceptors).

Ontology suggestions: HGNC:21653 (*TOPORS*); MONDO:0700233; parent disease MONDO:0019200 (retinitis pigmentosa); GO:0061630 (ubiquitin protein ligase activity); GO:0019789 (SUMO transferase activity).

Finding 2 — A distinctive early perivascular RPE atrophy phenotype (variable)

The original description noted **"a distinct phenotype ... at the earlier stages of the disease, with an unusual perivascular cuff of retinal pigment epithelium atrophy, which was found surrounding the superior and inferior arcades in the retina"** (PMID: 17924349). However, this feature is **not universal**: a subsequent report of two additional probands found that **"Contrary to previous reports, no distinct clinical phenotype was seen in these patients,"** who showed typical advanced adRP (PMID: 18509552).

Interpretation: Perivascular RPE atrophy may be an early, stage-dependent and variably expressed sign rather than a constant diagnostic marker. It should raise suspicion for *TOPORS* in an adRP pedigree, but its absence does not exclude the diagnosis.

Ontology suggestions: HP:0007894 (retinal pigment epithelial atrophy); HP:0000488 (retinopathy); HP:0000510 (rod-cone dystrophy).

Finding 3 — TOPORS is a photoreceptor connecting-cilium/basal-body protein; loss impairs outer-segment formation

TOPORS localizes to **the basal bodies of the photoreceptor connecting cilium and to centrosomes** of cultured cells, and is **"a novel component of the photoreceptor sensory cilium"** (PMID: 21159800). Functional confirmation came from zebrafish: **"Morpholino-mediated silencing of topors in zebrafish embryos demonstrates in another species a comparable retinal problem as seen in humans, resulting in defective retinal development and failure to form outer segments"** (PMID: 21159800). The retinal defect was rescuable by human *TOPORS* mRNA, confirming conservation.

Interpretation: This places RP31 within the broad class of **retinal ciliopathies**. The photoreceptor connecting cilium is the trafficking conduit between the inner segment (where proteins are synthesized) and the outer segment (the light-sensing compartment). Reduced TOPORS dosage impairs this trafficking, preventing proper outer-segment assembly/maintenance and leading to photoreceptor degeneration.

Ontology suggestions: GO:0060271 (cilium assembly); GO:0032391 (photoreceptor connecting cilium); GO:0120289 (photoreceptor outer segment disc assembly); CL:0000210 (photoreceptor cell); CL:0000604 (retinal rod cell); CL:0000573 (retinal cone cell); UBERON:0001789 (photoreceptor outer segment).

Finding 4 — TOPORS is a dual ubiquitin/SUMO1 E3 ligase (STUbL) with pleiotropic nuclear functions

Beyond the retina, TOPORS is a **RING-domain dual ubiquitin and SUMO1 E3 ligase** that localizes to PML nuclear bodies and functions as a **SUMO-targeted ubiquitin ligase (STUbL)**. It **"promotes DPC repair by functioning as a SUMO-targeted ubiquitin ligase (STUbL), combining ubiquitin ligase activity through its RING domain with poly-SUMO binding via SUMO-interacting motifs, analogous to the STUbL RNF4"** (PMID: 38649616). It regulates **p53** — it is **"an ubiquitin and small ubiquitin-like modifier ubiquitin-protein isopeptide ligase (SUMO E3) ligase"** (PMID: 19473992) — along with NKX3.1 and SUMOylated DNMT1 crosslinks. In mice, *Topors* deficiency produces genomic instability: **"heterozygotes were found to have an increased incidence of malignancy, involving a variety of tissues"** (PMID: 20429939).

Interpretation: TOPORS is a multifunctional, ubiquitously expressed protein. Its DNA-protein-crosslink-repair and p53-regulatory roles explain its emerging cancer biology (it modulates sensitivity to DNA hypomethylating agents in myeloid malignancy), but these nuclear functions are **largely distinct from — though mechanistically illuminating of — the retinal ciliary phenotype**. The post-mitotic photoreceptor is likely most sensitive to the loss of the ciliary/proteostatic pool of TOPORS rather than its genome-maintenance role.

Ontology suggestions: GO:0016567 (protein ubiquitination); GO:0016925 (protein sumoylation); GO:0006302 (double-strand break repair); GO:0016605 (PML body).

Finding 5 — Substantial quality-of-life burden and treatable complications

Although RP31 itself is non-fatal, it imposes a real burden. In Australian adults with inherited retinal diseases, **"Median EQ-5D-5L utility was 0.88 and 0.91 among people with widespread and central involvement, respectively"** — the lower utility (0.88) applies to widespread retinal disease such as RP (PMID: 38226448). Depression is an independent determinant of function: **"The RP patients with depression had poorer vision-related functions compared with those patients without depression, which cannot be explained by the visual acuity"** (PMID: 18356260). Posterior subcapsular cataract and cystoid macular edema are frequent RP complications; cataract surgery helps: **"Mean best-corrected visual acuity significantly improved after cataract surgery, from 1.09 ± 0.69 preoperatively to 0.614 ± 0.448 at 1 month postoperatively"** (PMID: 29217020), though RP surgery carries elevated risk of posterior capsular opacification, cystoid macular edema and zonular weakness (PMID: 39206084, PMID: 33489339).

Interpretation: Management should be holistic — not only ophthalmic (treating cataract and macular edema) but also addressing mental-health comorbidity, which independently worsens functional outcomes.

Finding 6 — TOPORS is a dual-inheritance ciliopathy gene

TOPORS causes disease through two genetically and clinically distinct routes. **Heterozygous truncating variants → non-syndromic adRP (RP31).** In contrast, **biallelic hypomorphic missense variants → recessive syndromic ciliopathy**: **"homozygosity for a TOPORS missense variant (c.29C > A; p.(Pro10Gln)) was identified in individuals with the ciliopathy oral-facial-digital syndrome in two families of Dominican descent"** and also in Joubert syndrome (PMID: 37227088). The p.Pro10Gln allele is a **Dominican founder allele** with high carrier frequency (identified in the BioMe biobank of 1880 Dominican individuals): the authors note their data **"nominates TOPORS as a novel causal gene for JBTS and suggests that TOPORS variants should be considered in the differential of ciliopathy-spectrum disease in individuals of Dominican ancestry"** (PMID: 37227088).

Interpretation: This is the unifying insight of the investigation. A gene-dosage/allele-type model explains the whole TOPORS disease spectrum:

Genotype	Allele effect	Residual function	Phenotype	Inheritance
Heterozygous truncating (LOF)	~50% dosage	Haploinsufficient in photoreceptors only	Retina-only adRP (RP31)	Autosomal dominant
Biallelic missense (e.g., p.Pro10Gln)	Hypomorphic	Low but non-zero, multisystem deficit	Syndromic ciliopathy (Joubert / OFD)	Autosomal recessive

Dual-inheritance behavior is now a recognized feature of a minority of IRD genes (PMID: 42321002).

Ontology suggestions: MONDO:0018772 (Joubert syndrome); MONDO:0009299 (orofaciodigital syndrome); GO:0060271 (cilium assembly).

Finding 7 — Retinal TOPORS interactome links it to ubiquitin-proteasome proteostasis at the photoreceptor/RPE interface

A yeast two-hybrid screen of a **human retinal cDNA library** identified the 26S proteasome regulatory subunit 4 (P26s4/PSMC1) as a TOPORS interactor: **"The screen led to isolation of the 26 S protease regulatory subunit 4 (P26s4/PSMC1), an ATPase indispensable for correct functioning of UPS-mediated proteostasis"** (PMID: 26872363). Crucially, **"TOPORS and P26s4 co-localise at the centrosome in cultured cells"** (PMID: 26872363), with strong P26s4 reactivity at the RPE/photoreceptor interface in mouse retina. Additional TOPORS substrates relevant to retinal homeostasis include DJ-1/PARK7 and APOBEC2; in zebrafish, Toporsa interacts with Ubc9 and Pou6f2, and SUMOylation regulates APOBEC2 localization during retina regeneration (PMID: 25190811).

Interpretation: This provides a **second, complementary retinal mechanism** beyond ciliary trafficking: TOPORS couples the ubiquitin-proteasome system to the centrosome/basal body. Photoreceptors have extraordinary protein turnover demands (outer-segment discs are shed and renewed daily), making them exquisitely sensitive to any reduction in proteostatic capacity. Reduced TOPORS dosage may therefore degrade both ciliary trafficking **and** protein quality control at the connecting cilium / RPE interface.

Ontology suggestions: GO:0043161 (proteasome-mediated ubiquitin-dependent protein catabolic process); GO:0005813 (centrosome); CL:0002586 (retinal pigment epithelial cell); UBERON:0001782 (retinal pigment epithelium).

Finding 8 — Adult-onset non-syndromic rod-cone dystrophy, confirmed across diverse cohorts

Recent cohorts reaffirm the clinical picture. In a UAE series of non-syndromic adult-onset rod-cone dystrophy (symptomatic at or after age 16), **"Autosomal dominant RCD was noted in two cases, related to TOPORS and PRPF31"** (PMID: 42627262). A 2024 case report described a novel *TOPORS* mutation with clinical findings in adRP (PMID: 38592336). A Belgian adRP cohort also identified a novel frameshift *TOPORS* mutation among splicing-factor-dominated adRP (PMID: 28076437). Together these confirm *TOPORS* as a recurrent, if rare, cause of **adult-onset dominant rod-cone dystrophy across diverse populations** (French-Canadian, German, US, Belgian, Chinese, Emirati).

Ontology suggestions: HP:0000510 (rod-cone dystrophy); HP:0007737 (bone spicule-like pigmentary changes); HP:0000662 (nyctalopia); HP:0001133 (constriction of peripheral visual field).

Finding 9 — Slow, quantifiable progression; weak evidence for nutritional therapy

RP (including RP31) progresses slowly but measurably. In X-linked RP, **"Using different clinical biomarkers, XLRP progresses at a rate of 4 to 19% per year"** (PMID: 33258268). Ellipsoid-zone (EZ) width on OCT is the most reliable structural biomarker: **"The ellipsoid zone line width declined with a yearly average rate of 76.4 µm (4.16%/year)"** in a sibling cohort (PMID: 29940899). Cone function by full-field flicker ERG declines ~18.7%/year (PMID: 33364083). For therapy, a Cochrane review is unequivocal: **"At this time, there is no proven therapy for RP"** (PMID: 32573764). DHA slowed some visual-field outcomes only in post-hoc/ancillary analyses in XLRP (PMID: 26469750). Progression is heterogeneous even between affected siblings (PMID: 29940899).

Interpretation: EZ width should be the primary structural endpoint for any future RP31 natural-history study or trial. No supplement has proven efficacy; vitamin A/DHA should be discussed as low-certainty, not standard-of-care.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

```
1. Heterozygous truncating variant in TOPORS (9p21.1), e.g. p.Glu808X, p.Arg857GlyfsX9
      │  leads to
      ▼
2. Nonsense-mediated decay / absence of mutant protein → ~50% reduction in
   functional TOPORS dosage (HAPLOINSUFFICIENCY; mutant protein not detected in patients)
      │  results in
      ▼
3. Reduced TOPORS at the photoreceptor connecting-cilium basal body / centrosome
      │
      ├──► BRANCH A (ciliary trafficking; DEMONSTRATED):
      │      impaired protein transport through the connecting cilium
      │          │ leads to
      │          ▼
      │      failure of photoreceptor OUTER-SEGMENT formation/maintenance
      │      (shown in zebrafish knockdown)
      │
      └──► BRANCH B (proteostasis; INFERRED for retina):
             reduced TOPORS–PSMC1(26S proteasome) coupling at the centrosome
             and RPE/photoreceptor interface
                 │ leads to
                 ▼
             impaired ubiquitin-proteasome protein quality control in cells
             with extreme protein-turnover demand
      │
      ▼  (branches converge)
4. Progressive PHOTORECEPTOR DEGENERATION (rods first, then cones)
      │  results in
      ▼
5. Rod dysfunction → nyctalopia (night blindness); sometimes early
   perivascular RPE atrophy around arcades
      │  leads to
      ▼
6. Progressive mid-peripheral then peripheral visual-field constriction;
   bone-spicule pigmentation; later cone involvement → central acuity loss
      │  leads to
      ▼
7. Secondary complications: posterior subcapsular cataract, cystoid macular
   edema → further visual impairment and reduced quality of life
```

Steps 1–2 and Branch A are **demonstrated** (human genetics + zebrafish). Branch B is **supported by protein-interaction/localization data but not yet proven to drive degeneration in vivo** — it is the strongest candidate second mechanism. Steps 5–7 are **clinically established** for RP generally and observed in RP31 cohorts.

Dual-inheritance dosage model

```
TOPORS functional dosage ──────────────────────────────────►
   0% ............ low (hypomorphic) ............ 50% ............ 100%
   |                      |                          |               |
 (lethal?)        BIALLELIC missense          HETEROZYGOUS         normal
                  p.Pro10Gln (Dominican       truncating LOF
                  founder) → RECESSIVE         → DOMINANT
                  SYNDROMIC ciliopathy         RETINA-ONLY adRP
                  (Joubert / OFD)              (RP31)
```

The retina is the most dosage-sensitive tissue (50% is enough to cause isolated retinopathy), whereas multisystem ciliopathy requires a deeper, biallelic reduction in function.

Upstream vs downstream summary

Level	Element	Status
Upstream trigger	*TOPORS* truncating variant, haploinsufficiency	Demonstrated (human)
Proximal molecular	Loss of basal-body/centrosomal TOPORS; impaired cilium trafficking	Demonstrated (localization + zebrafish)
Parallel molecular	Reduced UPS proteostasis (PSMC1/DJ-1/ APOBEC2)	Supported (interactome)
Cellular	Outer-segment failure → photoreceptor death	Demonstrated / inferred
Tissue	Rod > cone loss; RPE atrophy	Clinical
Organ	Progressive retinal degeneration	Clinical
Secondary	Cataract, CME, QoL/mental-health burden	Clinical

Section-by-Section Disease Characterization

1. Disease Information

Non-syndromic, adult-onset autosomal dominant retinitis pigmentosa caused by *TOPORS*. **Identifiers:** OMIM #609923 (RP31); MONDO:0700233; gene *TOPORS* HGNC:21653, NCBI Gene 10210, Ensembl ENSG00000197579; locus 9p21.1. **Synonyms:** Retinitis pigmentosa 31, RP31, TOPORS-related adRP, TOPORS-related rod-cone dystrophy. **Source type:** aggregated disease-level resources (OMIM, Orphanet) plus individual case reports/cohorts.

2. Etiology

Primary cause: genetic — heterozygous truncating (nonsense/frameshift/insertion/deletion) LOF variants in *TOPORS* (germline). **Genetic risk:** carrying one pathogenic *TOPORS* allele (autosomal dominant). No established environmental risk or protective factors specific to RP31; general RP care avoids excessive vitamin A toxicity. **Gene–environment interactions:** not characterized for RP31.

3. Phenotypes (suggested HPO terms with typical characteristics)

Phenotype	HPO	Type	Onset	Progression	Frequency
Nyctalopia (night blindness)	HP:0000662	Symptom	Adult	Progressive	Typical/ early
Peripheral field constriction	HP:0001133	Sign	Adult	Progressive	Typical
Rod-cone dystrophy	HP:0000510	Sign	Adult	Progressive	Defining
Bone-spicule pigmentation	HP:0007737	Sign	Adult	Progressive	Common
Perivascular RPE atrophy	HP:0007894	Sign	Early	—	Variable
Reduced/abolished ERG	HP:0000512	Lab	Adult	Progressive	Typical
Posterior subcapsular cataract	HP:0007787	Complication	Adult	Progressive	Common (RP)
Cystoid macular edema	HP:0011505	Complication	Adult	Fluctuating	Common (RP)

Severity: variable; **QoL impact:** EQ-5D-5L utility ~0.88 for widespread retinal involvement (PMID: 38226448); depression independently worsens function (PMID: 18356260).

4. Genetic / Molecular Information

Causal gene: *TOPORS* (OMIM *609507), 3 exons, 1045 aa, RING-domain dual ubiquitin/SUMO1 E3 ligase. Variant types: truncating — nonsense (p.Glu808X), frameshift (p.Arg857GlyfsX9), small insertions/deletions; classified pathogenic/likely pathogenic per ACMG. Allele frequency: rare/ absent in gnomAD for truncating adRP alleles. Origin: germline. Functional consequence: loss of function → haploinsufficiency. Dual-inheritance allele: biallelic missense p.Pro10Gln (c.29C>A), hypomorphic, Dominican founder → recessive ciliopathy (PMID: 37227088). Modifier genes / epigenetics / chromosomal abnormalities:** none established for RP31.

5. Environmental Information

No established environmental, lifestyle, or infectious contributors. The disease is monogenic/Mendelian.

6. Mechanism / Pathophysiology

See the ordered causal chain and mechanistic model above. Key pathways: cilium assembly (GO:0060271), ubiquitin/SUMO conjugation (GO:0016567, GO:0016925), proteasomal degradation (GO:0043161). Cell types: photoreceptors (CL:0000210; CL:0000604 rod; CL:0000573 cone), RPE (CL:0002586). Subcellular: connecting cilium/basal body (GO:0032391), centrosome (GO:0005813), PML body (GO:0016605).

7. Anatomical Structures Affected

Primary organ: eye/retina (UBERON:0000970 eye; UBERON:0000966 retina). **Tissue/cell:** neural retina photoreceptor layer, RPE; rods affected before cones. **Subcellular:** photoreceptor outer segment (UBERON:0001789), connecting cilium, centrosome/basal body. **Localization:** bilateral, typically symmetric; early perivascular RPE atrophy around superior/inferior vascular arcades.

8. Temporal Development

Onset: adult (symptomatic at/after ~age 16); insidious, chronic. **Course:** slowly progressive, lifelong, no remission. **Progression rate:** EZ width ~4.2%/year; cone ffERG ~18.7%/year; overall biomarker decline ~4–19%/year (PMID: 33258268, PMID: 29940899, PMID: 33364083); heterogeneous even between siblings.

9. Inheritance and Population

Inheritance: autosomal dominant (truncating LOF). **Penetrance:** high but may be age-dependent; **expressivity:** variable. **Frequency:** ~1% of adRP (PMID: 18509552); overall RP prevalence ~1 in 3,700–4,000 (RP31 is a small fraction of that). **Founder effect:** p.Pro10Gln in Dominican ancestry (recessive syndromic form). **Sex ratio:** no strong predilection; autosomal. No anticipation or repeat-expansion mechanism.

10. Diagnostics

Genetic testing is definitive: WES/WGS or targeted adRP/IRD NGS gene panels including *TOPORS*; single-gene testing after a suggestive pedigree. **Clinical tests:** full-field ERG (reduced/abolished rod then cone responses), OCT (EZ width — key biomarker), fundus autofluorescence, visual fields (progressive constriction), fundus exam (bone spicules, perivascular RPE atrophy). **Differential diagnosis:** other adRP genes — *RHO*, *RP1*, *PRPH2*, splicing factors *PRPF31/PRPF8/SNRNP200* (PMID: 28076437), *NR2E3*, *PROM1*; and distinguishing syndromic ciliopathies when biallelic *TOPORS* variants are found.

11. Outcome / Prognosis

Not life-limiting (no mortality). **Morbidity:** progressive, irreversible vision loss leading to legal blindness in many over decades; EZ loss and ERG decline predict course. **Complications:** cataract (PSC), cystoid macular edema, posterior capsular opacification. **Recovery:** none spontaneously; cataract surgery can improve acuity (logMAR 1.09→0.61) (PMID: 29217020). **Prognostic biomarkers:** EZ width, ffERG amplitude.

12. Treatment

No approved disease-specific therapy — "there is no proven therapy for RP" (PMID: 32573764). **Supportive:** low-vision aids, genetic counseling. **Complication management:** phacoemulsification + IOL for cataract (NCIT: Phacoemulsification); carbonic anhydrase inhibitors / anti-VEGF considered for CME. **Nutritional:** vitamin A/DHA — low-certainty, not standard-of-care; DHA benefit only in post-hoc analyses (PMID: 26469750). **Experimental/future:** because the mechanism is LOF/haploinsufficiency, RP31 is theoretically amenable to **gene augmentation** (AAV-*TOPORS*); antisense/RNA approaches are advancing for other adRP genes (e.g., *PRPF31*, *RHO*) but none exists for *TOPORS* yet.

13. Prevention

No primary prevention (genetic disease). **Secondary/tertiary:** regular ophthalmic monitoring; early cataract/CME management. **Genetic counseling** (50% transmission risk, autosomal dominant); carrier/cascade testing; prenatal/preimplantation genetic testing available for known familial variants. For Dominican-ancestry families, counsel on the recessive syndromic p.Pro10Gln allele.

14. Other Species / Natural Disease

Orthologs: zebrafish *toporsa/toporsb* (retinal model), mouse *Topors* (Gene 106021; KO → genomic instability/malignancy, PMID: 20429939), *Drosophila dtopors* (meiotic/nuclear-structure role, PMID: 21900273). No well-documented naturally occurring companion-animal RP31 equivalent; mechanism conserved across vertebrates. Not zoonotic.

15. Model Organisms

Zebrafish morpholino knockdown — best disease-relevant model; recapitulates outer-segment failure, rescuable by human mRNA (PMID: 21159800). **Mouse** *Topors* knockout — models the genomic-instability/tumor role, not the retinal phenotype directly (PMID: 20429939). **Cellular:** hTERT-RPE1 and 661W cells used for TOPORS/PSMC1 centrosomal localization (PMID: 26872363). **Limitation:** no published knock-in mouse replicating human haploinsufficiency adRP; a *Topors*+/− retinal-aging model is a clear gap.

Evidence Base

PMID	Title (abbreviated)	Supports / Role
17924349	*Mutations in TOPORS cause adRP with perivascular RPE atrophy*	Establishes *TOPORS*/9p21.1 as RP31 gene; haploinsufficiency; perivascular phenotype
18509552	*Mutations in TOPORS cause 1% of adRP*	Quantifies contribution; variable phenotype
21159800	*TOPORS … is a cilia-centrosomal protein*	Ciliary localization; zebrafish outer-segment failure
26872363	*TOPORS interacts with PSMC1 (26S subunit)*	Retinal proteostasis mechanism
37227088	*TOPORS as a novel causal gene for Joubert syndrome*	Dual inheritance; recessive syndromic ciliopathy; Dominican founder allele
38649616	*TOPORS/RNF4 STUbL activities*	Defines STUbL / DPC-repair molecular function
19473992	*Plk1-mediated phosphorylation of Topors regulates p53*	Dual ligase; p53 regulation
20429939	*Topors deficiency → genetic instability/malignancy in mice*	In vivo tumor-suppressor function; mouse KO
42627262	*Non-syndromic adult-onset rod-cone dystrophy*	Confirms TOPORS in recent AD RCD cohort
38592336	*adRP secondary to TOPORS: novel mutation*	Recent clinical confirmation
28076437	*Splicing-factor genes in Belgian adRP*	Novel TOPORS frameshift; differential-diagnosis context
38226448	*EQ-5D-5L utilities in IRD*	QoL burden
18356260	*Depression and QoL in RP*	Mental-health comorbidity
29217020	*Cataract surgery outcomes in retinal dystrophy*	Treatable complication
39206084 / 33489339	*Cataract surgery in RP*	Surgical risks/benefits

PMID	Title (abbreviated)	Supports / Role
32573764	*Cochrane: vitamin A / fish oils for RP*	No proven therapy
26469750	*DHAX trial (DHA in XLRP)*	Weak/ancillary nutritional evidence
33258268 / 29940899 / 33364083	*RP natural history / biomarkers*	Progression rates (EZ, ffERG)
25190811	*Apobec2 / Toporsa in zebrafish retina regeneration*	TOPORS–SUMO–APOBEC2 retinal axis
42321002	*Dual-inheritance IRD genes*	Context for dual-inheritance behavior
21900273	*dTopors in Drosophila meiosis*	Conservation; nuclear-structure role

Evidence source distribution: human clinical genetics (multiple cohorts/case reports), in vitro/cell biology (yeast two-hybrid, co-IP, localization), model organism (zebrafish, mouse, Drosophila), and systematic review/Cochrane for therapy.

Limitations and Knowledge Gaps

1. **No RP31-specific natural-history study.** Progression rates cited are from RP broadly (largely XLRP/*USH2A*); gene-specific RP31 kinetics are unknown.
2. **Branch B (proteostasis) is not proven in vivo.** The TOPORS–PSMC1 link rests on interaction/localization data; whether reduced proteostasis drives photoreceptor death in a *TOPORS*+/− retina is untested.
3. **No haploinsufficiency mouse model** replicating the adRP phenotype; existing KO models the genome-instability/cancer role instead.
4. **Small patient numbers.** RP31 is ~1% of adRP; genotype–phenotype correlation (e.g., which truncating positions cause perivascular atrophy) is underpowered.
5. **Penetrance/expressivity not precisely quantified** for *TOPORS* adRP.
6. **No targeted therapy or trial** exists for *TOPORS*; gene-augmentation feasibility (promoter, dosage, AAV tropism to photoreceptors) is unexplored.

7. **Dual-inheritance thresholds** (how much residual function separates retina-only from syndromic disease) are inferred, not measured.

Proposed Follow-up Experiments / Actions

1. **Generate a *Topors* heterozygous / photoreceptor-conditional mouse** and perform longitudinal OCT/ERG to test whether haploinsufficiency alone causes age-dependent rod-cone degeneration, and to directly test the proteostasis (Branch B) hypothesis.
2. **Patient-derived iPSC retinal organoids** carrying truncating *TOPORS* alleles to quantify outer-segment formation, ciliary trafficking, and proteasome function; test AAV-*TOPORS* rescue.
3. **RP31-specific natural-history registry** using EZ width (OCT) as the primary structural endpoint and ffERG as functional endpoint, to establish gene-specific progression rates for future trial design.
4. **Deep genotype–phenotype study** across all reported *TOPORS* families to test whether perivascular RPE atrophy correlates with variant position/type, and to estimate penetrance/expressivity.
5. **Structure–function dosage experiments** on the p.Pro10Gln hypomorph vs truncating alleles to quantify the residual-activity threshold separating retina-only from syndromic ciliopathy.
6. **Gene-augmentation proof-of-concept:** optimize AAV-*TOPORS* vector/promoter for photoreceptor dosage; because the mechanism is LOF, augmentation (rather than suppression/editing) is the rational strategy.
7. **Carrier-screening guidance for Dominican-ancestry populations** given the p.Pro10Gln founder allele, to inform recessive-ciliopathy counseling.

Report compiled from 9 confirmed findings across 5 investigation iterations and 48 reviewed papers. Evidence spans human clinical genetics, cell biology, and model-organism studies. All mechanistic and clinical claims are cited to primary literature by PMID.
